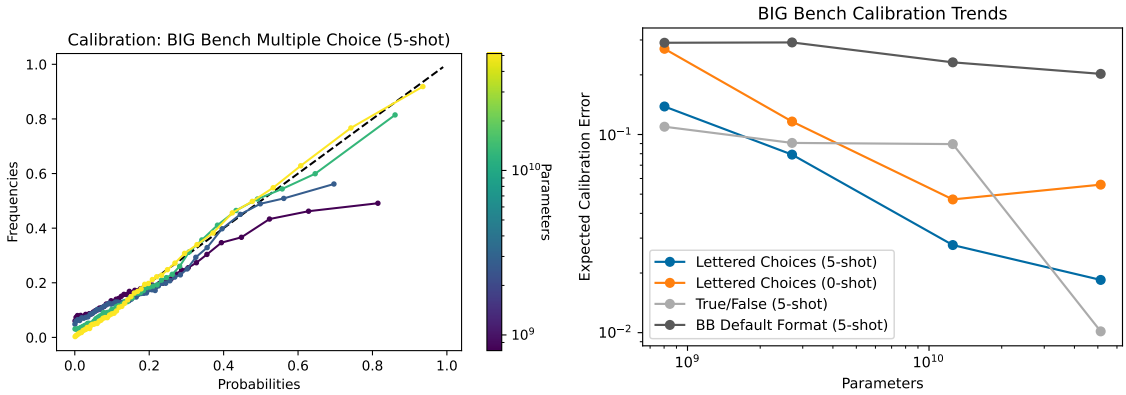


Figure 4 **(left)** We show calibration curves for various model sizes on all of the multiple choice tasks in BIG Bench, in the format described in section 2. We include a dashed line indicating perfect calibration. **(right)** Here we show trends in the expected calibration error on BIG Bench, for both multiple choice and a separate True/False format (see Section 3.2). We show the RMS calibration error in Figure 21 in the appendix.

Truthfulness [Evans et al., 2021] has been a recent focus of various works, including benchmarks [Lin et al., 2021] and the incorporation of web search and citation [Nakano et al., 2021, Menick et al., 2022] into language models. That said, truthfulness focuses primarily on factual accuracy "in the world", rather than on self-knowledge, or eliciting latent knowledge [Christiano et al., 2021]. We use "honesty" [Askell et al., 2021] as an umbrella term for ideas including truthfulness, calibration, self-knowledge, explainability, and non-deceptiveness. Language models finetuned to perform non-language tasks [Ahn et al., 2022, Dinh et al., 2022] might provide an interesting test-bed for honesty in the future.

Perhaps the work most similar to ours is [Mielke et al., 2020], which is a very interesting application of metacognition/self-evaluation to improve natural language calibration. Another quite similar work is the very recent [Lin et al., 2022], where the authors train language models to express their calibration on arithmetic in words, and also study a signal analogous to P(True).

2 Larger Models are Calibrated on Diverse Multiple Choice Questions

A model makes calibrated predictions if the probability it assigns to outcomes coincides with the frequency with which these outcomes actually occur. Language models are known to produce calibrated token-level probabilities. In this section we will see that language models can produce well-calibrated probabilities when they are asked to choose the correct answer from among several explicit options. We believe calibration is interesting on its own, but it is especially relevant to honesty, since a model that can produce calibrated answers to meta-questions like 'do you know the answer to X?' must know something about what it knows. Generally we can use calibration as a way to bootstrap towards self-knowledge.

We find that when multiple choice problems are formatted in this way (as used by e.g. [Rae et al., 2021]):

```
Question: Who was the first president of the United States?
Choices:
 (A) Barack Obama
 (B) George Washington
 (C) Michael Jackson
Answer:
```

and we identify the answers only by their labels, as e.g. ' (B)', our largest models tend to produce a well-calibrated probability distribution among the available options. We show the calibration chart for all multiple choice BIG Bench tasks, in this format, in Figure 4. As can be seen in Figure 6, models are well-calibrated (in this format) even for somewhat adversarial datasets like TruthfulQA[3] [Lin et al., 2021], as well as for QuALITY [Pang et al., 2021] and LogiQA [Liu et al., 2020].

[3]Though note that in this format, where the model sees its options before making a choice, we do not observe anti-scaling with model size.

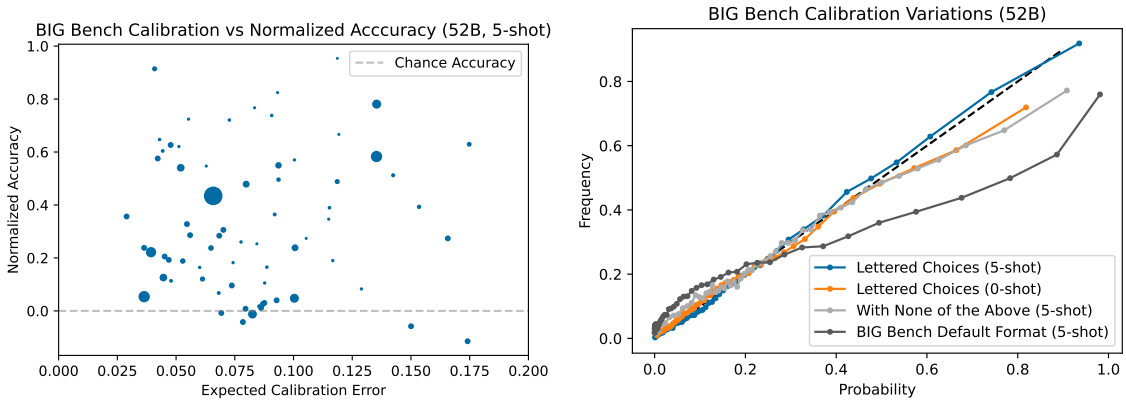


Figure 5 **(left)** We show expected calibration error versus normalized accuracy for all BIG Bench tasks; the number of problems in each task is represented by the marker sizes. We do not find any noticeable correlation between accuracy and calibration within BIG Bench. To normalize accuracies we linearly map chance accuracy to 0, keeping perfect accuracy at 1. **(right)** We compare calibration for several variations on BIG Bench evaluations: we vary between 0-shot and 5-shot, replace an answer option with "none of the above", and compare our format with letter-labeled choices to the default BIG Bench formatting.

It is crucial that the model gets to see the answer choices explicitly before choosing amongst them; without this, we would not expect a calibrated response, due to ambiguities and degeneracies among possible paraphrases and specializations of any given answer option (e.g. "Washington" vs "George Washington, the first US president"). As can be seen in figure 5, task formatting is important for achieving excellent calibration, and calibration improves as we pass from 0-shot to 5-shot evaluation. We expect calibration is also easier to achieve with this format because each answer option corresponds to a single token (this isn't the case in BIG Bench by default, see appendix A.4).

To simplify the interpretation of results, we reduce calibration curves to a single number by computing the expected calibration error (ECE), after binning the predictions in 10 equally-represented bins. On the right of Figure 4 we show scaling trends for calibration on BIG Bench. We typically find good calibration 0-shot, without any other prompt, though calibration improves as we pass from 0-shot to few-shot. We find that in almost all cases, calibration improves with model size and capability. Accuracy on the tasks also improves with model size, but as can be seen in Figure 5, we do not observe any obvious causal connection between accuracy and calibration. For details on how we obtain calibration charts and ECE see Appendix A.

In what follows we will work to leverage these calibration results to ask language models to evaluate what they do and do not know.

3 From Calibration to Knowing What You Know

If language models can answer multiple choice questions in a calibrated way, then one might hope that they can apply this ability to evaluate their own outputs. That is, we can simply ask models to generate potential answers to questions, and then have the model evaluate whether any of them are correct. In this section we will begin to explore this idea by reformulating existing tasks. Then in section 4 we will study self-evaluation.

3.1 Replacing an Option with 'None of the Above' Harms Performance and Calibration

We have seen that language models can produce calibrated probabilities for multiple choice questions, at least when the questions and choices are provided in the right format. However, to achieve this feat the model only needs to determine the relative weight for several concrete options, when they are compared to each other.

We are more interested in whether the model actually knows whether each of the answer options is correct, when judged independently. To probe this question, we modified our multiple choice evaluations by replacing their final option with "none of the above". This procedure can make questions that do not actually have a unique correct answer ambiguous or impossible, but for many tasks it should result in a well-defined new evaluation. In particular this procedure appears to be sensible for the vast majority of questions in MMLU. Concretely this means that we took questions such as the example in section 2 and replaced them with:

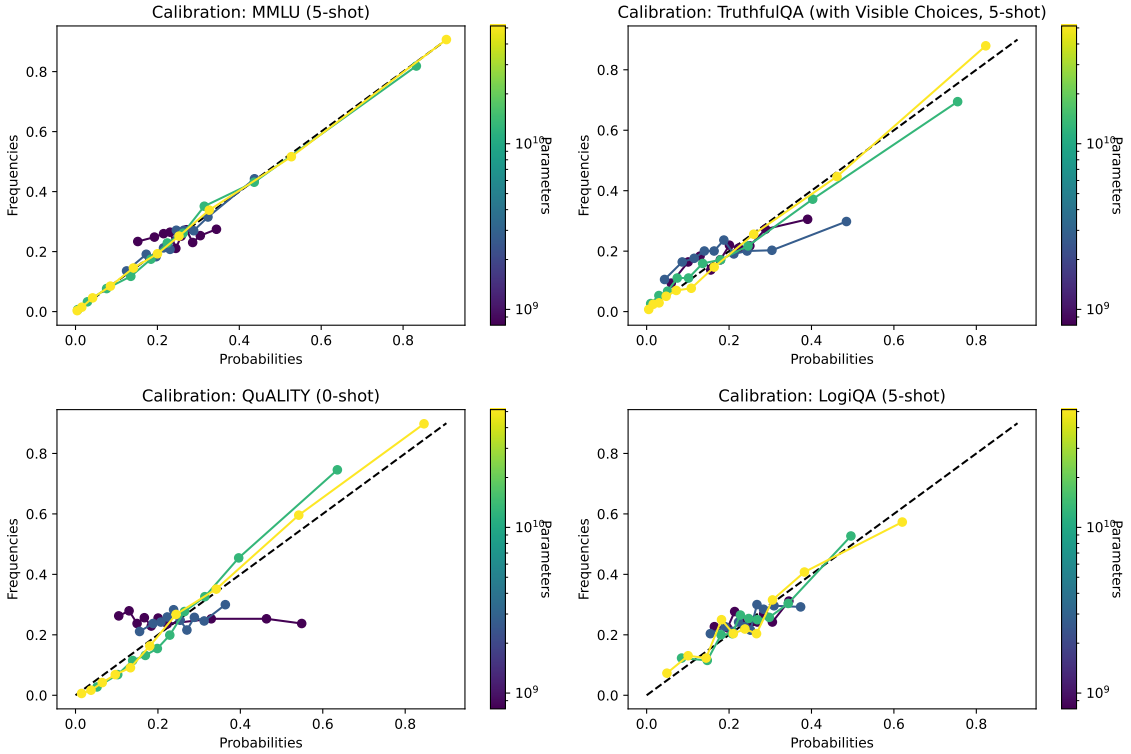


Figure 6 We show calibration curves for various model sizes on MMLU, TruthfulQA, QuALITY, and LogiQA in the format described in section 2. We emphasize that even on tasks that are difficult for LMs, like LogiQA, and on a somewhat adversarial evaluation like TruthfulQA, large models can achieve good calibration when the available answer options are presented to them in a favorable format. We include a dashed line indicating perfect calibration.

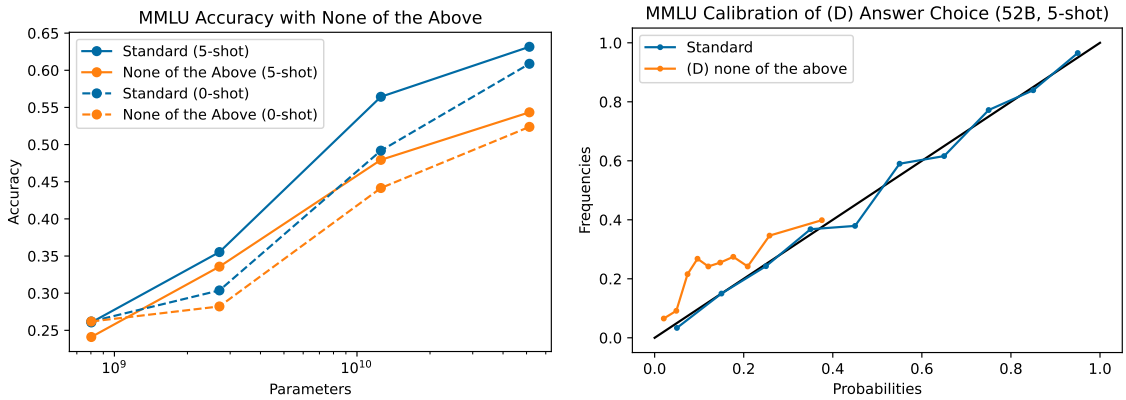


Figure 7 **(left)** We show accuracy on MMLU in the standard format, and after replacing option (D) with "none of the above". This replacement decreases accuracy very significantly. **(right)** We show calibration specifically for the (D) answer option, in the standard form of MMLU and with (D) as "none of the above". The latter makes calibration much worse, and in particular the model seems strongly biased against using this option, which also harms accuracy.